Total No. of Printed Pages: 1

SUBJECT CODE NO: RNP-8006
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III
Examination May/June-2026
Data Structures

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from question no's 2, 3, 4, 5, 6 and 7

SECTION-A

Q.1 Solve any TEN from the following questions: **20**

a) Define array. Give two examples.
b) What is pointer?
c) Define Big-O notation.
d) Write two applications of a queue.
e) What is a dequeue?
f) Define doubly linked list.
g) What is a binary tree?
h) What is Binary Search Tree (BST)?
i) State two differences between insertion sort and selection sort.
j) What is hashing?
k) Define graph.
l) What is BFS traversal?
m) Define heap sort.

SECTION-B

Q.2	a)	Explain different data structure operations with suitable examples.	**05**
	b)	Explain stack using array and its operations.	**05**
Q.3	a)	Explain circular queue with example.	**05**
	b)	Write algorithm for insertion and deletion in linked list.	**05**
Q.4	a)	Explain binary tree representation in memory.	**05**
	b)	Explain tree traversal techniques.	**05**
Q.5	a)	Explain merge sort with example.	**05**
	b)	Compare different sorting techniques.	**05**
Q.6	a)	Explain linear and binary search.	**05**
	b)	Explain hashing and hash functions.	**05**
Q.7	a)	Explain DFS traversal with example.	**05**
	b)	Explain Kruskal's algorithm.	**05**